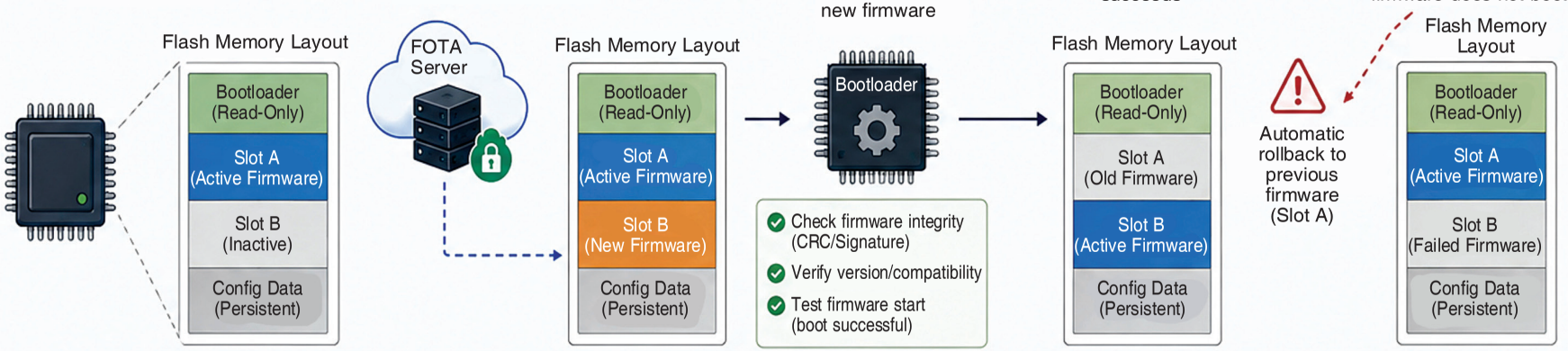
Reliable Firmware Storage and Recovery Mechanisms

A/B PARTITIONING ARCHITECTURE (RECOMMENDED)

1. Current running firmware in Slot A
2. New firmware downloaded into Slot B
3. Device reboots. Bootloader verifies new firmware
4. If verification succeeds
5. If verification fails or firmware does not boot



Report failed updates to the cloud
Devices automatically report rollback events so operators can identify affected devices and investigate the root cause.

Visibility into update failures | Identify affected devices | Investigate and resolve issues

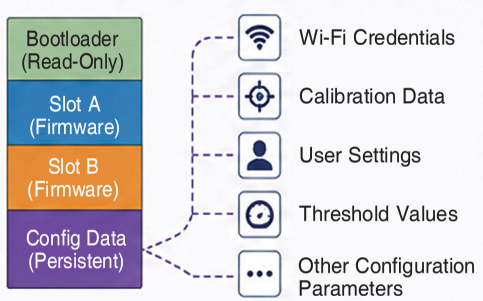
BOOTLOADER: THE GUARDIAN



Bootloader (Stored in protected area e.g., LDROM)

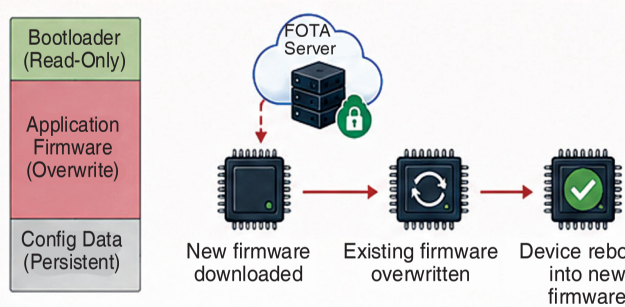
- Resides in a reserved flash segment (read-only)
- Not modified during regular firmware updates
- Verifies firmware before execution
- Ensures safe recovery in case of app update failure

PRESERVE CONFIGURATION DATA



Configuration partition remains intact across firmware updates, so the device does not need to be reconfigured.

SINGLE-PARTITION UPDATE (FOR MEMORY-LIMITED DEVICES)



Lower memory usage, but higher risk:
If update fails or is interrupted, device may become unusable (no automatic rollback).

LEGEND: → Process Flow | - - → Data/Connection | ✓ Success/Verified | ⚠ Failure/Risk